

Lecture 21 Notes: Cost of Capital and Corporate Valuation

Corporate Finance

I. Stock Valuation: Dividend Discount Model (DDM)

Key Principle

To find the value of any stock, we calculate its **Intrinsic Value**. The value today is the **Present Value of expected future dividends**.

Types of Growth Models

Model	Description
Zero Growth	Dividends are constant (perpetuity)
Constant Growth (Gordon)	Dividends grow at constant rate forever
Multistage Growth	Supernormal growth initially, then constant

II. Formulas for Growth and Required Return

Growth Rate Formula

$$g = ROE \times b$$

Where:

- ROE = Return on Equity
- b = Retention Ratio = $1 - \text{Payout Ratio}$
- $\text{Payout Ratio} = \frac{\text{Dividends per Share}}{\text{Earnings per Share}}$

Required Return (r_s) - Two Methods

Method 1: CAPM (Capital Asset Pricing Model)

$$r_s = r_{RF} + \beta \times (r_m - r_{RF})$$

Method 2: DDM (Dividend Discount Model)

$$r_s = \frac{D_1}{P_0} + g$$

Where:

- D_1/P_0 = Dividend Yield
- g = Growth Rate (Capital Gains Yield)

III. Practice Problems - Stock Valuation

Problem 1: Constant Growth DDM

Given:

$$D_1 = \$1.80 \quad (\text{expected dividend})$$

$$g = 6\% = 0.06$$

$$r_s = 12\% = 0.12$$

Solution:

$$\hat{P}_0 = \frac{D_1}{r_s - g} = \frac{1.80}{0.12 - 0.06} = \frac{1.80}{0.06} = \$30$$

Answer

$$P_0 = \$30$$

Problem 2: Finding Price Next Year

Given:

$$P_0 = \$38$$

$$D_0 = \$2.00$$

$$g = 5\% = 0.05$$

Step 1: Calculate expected dividend (D_1)

$$D_1 = D_0 \times (1 + g) = 2.00 \times 1.05 = \$2.10$$

Step 2: Calculate required return (r_s)

$$r_s = \frac{D_1}{P_0} + g = \frac{2.10}{38} + 0.05 = 0.0553 + 0.05 = 0.1053 = 10.53\%$$

Step 3: Calculate price one year from now (P_1)

$$P_1 = P_0 \times (1 + g) = 38 \times 1.05 = \$39.90$$

Answer

$$P_1 = \$39.90$$

$$r_s = 10.53\%$$

Problem 3: Non-Constant (Supernormal) Growth

Given:

$$\begin{aligned}D_0 &= \$2.75 \\g_1 &= 18\% \text{ for 2 years} \\g_c &= 6\% \text{ thereafter (constant)} \\r_s &= 12\% = 0.12\end{aligned}$$

Step 1: Calculate dividends for supernormal period.

$$\begin{aligned}D_1 &= D_0 \times (1 + g_1) = 2.75 \times 1.18 = \$3.245 \\D_2 &= D_1 \times (1 + g_1) = 3.245 \times 1.18 = \$3.829\end{aligned}$$

Step 2: Calculate the first dividend of constant growth period.

$$D_3 = D_2 \times (1 + g_c) = 3.829 \times 1.06 = \$4.058$$

Step 3: Calculate the price at Year 2 (P_2).

$$P_2 = \frac{D_3}{r_s - g_c} = \frac{4.058}{0.12 - 0.06} = \frac{4.058}{0.06} = \$67.63$$

Step 4: Discount all cash flows to present.

$$\begin{aligned}\hat{P}_0 &= \frac{D_1}{(1.12)^1} + \frac{D_2 + P_2}{(1.12)^2} \\&= \frac{3.245}{1.12} + \frac{3.829 + 67.63}{1.2544} \\&= 2.897 + \frac{71.459}{1.2544} \\&= 2.897 + 56.97 = \$59.87\end{aligned}$$

Answer

$$\hat{P}_0 = \$59.87$$

IV. The Corporate Valuation Model (Free Cash Flow)

When to Use Corporate Valuation

For **startups or growth stocks** that do not pay dividends in their initial years, we use the **Corporate Valuation Model** based on **Free Cash Flows (FCF)**.

Free Cash Flow (FCF) Formula

$$FCF = [EBIT \times (1 - T)] + \text{Depreciation} - \text{Capital Expenditures} - \Delta \text{Net Operating Working Capital}$$

Where:

- $EBIT(1 - T) = \text{After-tax operating income}$
- Depreciation is added back because it is a **non-cash charge**
- $\text{CapEx} = \text{Investment in fixed assets}$
- $\Delta NOWC = \text{Change in Net Operating Working Capital}$

Firm Value

$$V_{Firm} = \sum_{t=1}^{\infty} \frac{FCF_t}{(1 + WACC)^t}$$

Constant Growth FCF Model

$$V_{Firm} = \frac{FCF_1}{WACC - g}$$

Equity Value per Share

$$V_{Equity} = V_{Firm} - V_{Debt}$$

$$\hat{P}_0 = \frac{V_{Equity}}{\text{Shares Outstanding}}$$

V. Cost of Preferred Stock

Preferred Stock Characteristics

Preferred stock is a **hybrid security** with features of both stocks and bonds:

- Pays a **fixed dividend**
- Preference over common stockholders for dividends and liquidation
- Paid after debt holders

Cost of Preferred Stock Formula

$$r_p = \frac{D_p}{P_p}$$

Where:

- $r_p = \text{Cost of preferred stock}$
- $D_p = \text{Preferred dividend (fixed)}$
- $P_p = \text{Current price of preferred stock}$

VI. Cost of Debt and the Tax Shield

Key Principle

Debt is **tax-deductible**, which provides a **tax shield**. This makes the actual cost of debt lower for the firm.

After-Tax Cost of Debt Formula

$$r_d(1 - T)$$

Where:

- r_d = Before-tax cost of debt (interest rate)
- T = Marginal tax rate

Example

Loan Amount = \$100,000

Interest Rate = 8%

Interest Expense = \$100,000 $\times$ 0.08 = \$8,000

Tax Rate(T) = 30% = 0.30

Tax Saving = \$8,000 $\times$ 0.30 = \$2,400

After-Tax Interest Cost = \$8,000 $-$ \$2,400 = \$5,600

VII. Weighted Average Cost of Capital (WACC)

WACC Formula

$$WACC = w_d \times r_d(1 - T) + w_p \times r_p + w_s \times r_s$$

Where:

- w_d = Weight of debt
- w_p = Weight of preferred stock
- w_s = Weight of common equity
- $w_d + w_p + w_s = 1$

Key Insight

WACC is the average cost of capital for all sources used by the company.

Formula Summary Sheet

Key Formulas

$$\text{Growth Rate : } g = ROE \times b \quad (1)$$

$$\text{CAPM : } r_s = r_{RF} + \beta(r_m - r_{RF}) \quad (2)$$

$$\text{DDM Required Return : } r_s = \frac{D_1}{P_0} + g \quad (3)$$

$$\text{Constant Growth DDM : } \hat{P}_0 = \frac{D_1}{r_s - g} \quad (4)$$

$$\text{Zero Growth DDM : } \hat{P}_0 = \frac{D}{r_s} \quad (5)$$

$$\text{Free Cash Flow : } FCF = EBIT(1 - T) + \text{Depreciation} - \text{CapEx} - \Delta\text{NOWC} \quad (6)$$

$$\text{Firm Value (Constant Growth) : } V_{Firm} = \frac{FCF_1}{WACC - g} \quad (7)$$

$$\text{Equity Value per Share : } \hat{P}_0 = \frac{V_{Firm} - V_{Debt}}{\text{Shares Outstanding}} \quad (8)$$

$$\text{Cost of Preferred Stock : } r_p = \frac{D_p}{P_p} \quad (9)$$

$$\text{After-Tax Cost of Debt : } r_d(1 - T) \quad (10)$$

$$\text{WACC : } WACC = w_d r_d(1 - T) + w_p r_p + w_s r_s \quad (11)$$